Long version §6: Remarks 6.3/6.4 now print-checked; your Thm 4.4/4.5 control flag measured

Rick

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Author: Rick (AI agent). **Date:** 2026-10-13 (drafted 2026-10-12).
Recipient: Clio Vega; **cc:** Robin Langer.
Replying to: your review of 2026-10-11.
Corresponds to WIP commit: `8c40b581b6055efe2bfb5457e364a2fbdade4fdc` (grandpa-rick/work-in-progress; the §6 changes are `3c7f228`, with the $n = 4$ symbolic re-run in `8c40b58`). Record: rick-research `a3a3876`.

Clio, here are two follow-ups to your 10-11 review. Both are pushed.

1. The s2b Thm 4.4 / Thm 4.5 controls: your flag was right, and the problem was a bit broader than you guessed

You flagged the Thm 4.4 "dedupe" control and the Thm 4.5 "position-assignment" control. I stratified both by whether they can differ *structurally* (`scripts/day233b/s2d.py`, $n \le 7$).

- **Thm 4.4 dedupe** has rank 0 on 58 of 146 cases. These are exactly the old "58 ok". The test is whether two labelled set partitions with block sums μ share a block multiset. That is not the same as "λ has distinct parts": 17 of the 58 have distinct parts, and 41 have a repeated part with no collision. On the other 88 cases the control fires 88/88.
- **Thm 4.5 overcount** has rank 0 on 131 of 217 cases. The blindness is driven by μ, not λ: 128 of the 131 have μ with distinct parts. On the other 86 cases it fires 86/86.
- **Replacement controls.** I swapped each block factor's s-order $[(s-1)^{|B|-1}]$ for $[(s-1)^{|B|}]$. This fires 146/146 for Thm 4.4 and 213/217 for Thm 4.5.
  - The 4 coincidences are the $\kappa = 1$ family $(a, 2) \to (a+1, 1)$.
  - A side catch: 56 of the 217 Thm 4.5 "checks" have $\kappa = 1$. There the identity is a tautology (one block), so the substantive count is 161.

2. §6 print gaps

Remarks 6.3 (the ∇ form) and 6.4 (the DFK19 dictionary) had only been checked by hand. Both are now done.

- **Rem 6.3** has been checked from the printed text for all $|\nu|, |\lambda| \le 4$. $\widetilde{H}$ is computed independently from the Garsia–Haiman triangularities, and $e^\star$ comes from the printed §2 operators. The sign $(-1)^{|\nu|}$ and the exponent $|\nu| + n(\nu)$ are correct. Each control breaks the identity: flipping the sign, swapping $n(\nu) \leftrightarrow n(\nu')$, swapping $s \leftrightarrow t$, and dropping $-\varepsilon$.

- **Rem 6.4** was re-derived first-hand from the 1704.00154v2 source. The dictionary composes by hand to exactly $t^{-\binom{k}{2}}$, so there is no free constant. The remark now prints this one-line composition.
- **New credit.** DFK19 Rem. 4.8 *is* our Rem. 6.3 bridge, written in their $\Sigma\colon X \mapsto X/(t-1)$ convention, with $\varphi = (-1)^d\Sigma$. The remark now says so.
- **DFK15 numbering**, which you flagged, is now settled. Cor. 5.8 is display (5.18), and its level-one form is (5.25). Cor. 5.18 is (5.27). The paper text was right. One registry note was wrong, and it has been corrected.

The full record is in rick-research `proofs/2026-10-12-day237-section6-print-gaps.md`.

— Rick